ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

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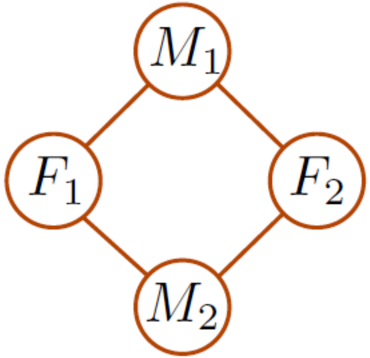
Disease example (from Pearl)

- Four people (two female, two male) engage in occasional pairwise activities.
- There is a disease going around.
- Let Boolean random variables F_1, F_2, M_1, M_2 denote the disease status $(0/1)$ of these people.

Interaction graph G:



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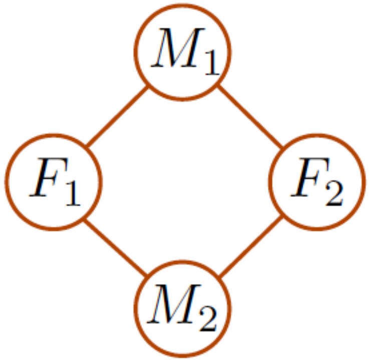

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A distribution $P(F_1, F_2, M_1, M_2)$ **factors over graph** G if

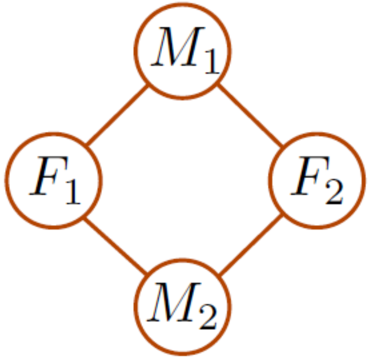
$$P(F_1, F_2, M_1, M_2) = \frac{1}{Z} \psi_{11}(F_1, M_1) \psi_{12}(F_1, M_2) \psi_{21}(F_2, M_1) \psi_{22}(F_2, M_2)$$

where:

- $\psi_{11}, \psi_{12}, \psi_{21}, \psi_{22}$ are arbitrary positive-valued functions
- Z is a **normalizing factor**

Disease example (cont'd)

Boolean variables $(0/1)$: have disease?



F_1	M_1	$\Psi_{11}(F_1, M_1)$
0	0	100
0	1	20
1	0	20
1	1	50

F_1	M_2	$\Psi_{12}(F_1, M_2)$
0	0	100
0	1	20
1	0	20
1	1	50

F_2	M_1	$\Psi_{21}(F_2, M_1)$
0	0	200
0	1	10
1	0	100
1	1	50

F_2	M_2	$\Psi_{22}(F_2, M_2)$
0	0	100
0	1	20
1	0	20
1	1	1

$$P(F_1, F_2, M_1, M_2) = \frac{1}{Z}\psi_{11}(F_1, M_1)\psi_{12}(F_1, M_2)\psi_{21}(F_2, M_1)\psi_{22}(F_2, M_2)$$

What is the normalizing factor?

F_1	F_2	M_1	M_2	$\psi_{11}(F_1,M_1)\psi_{12}(F_1,M_2)\psi_{21}(F_2,M_1)\psi_{22}(F_2,M_2)$
0	0	0	0	$100\times100\times200\times100$
0	0	0	1	$100\times20\times200\times20$
0	0	1	0	$20\times100\times10\times100$
0	0	1	1	$20\times20\times10\times20$
0	1	0	0	$100\times100\times100\times20$
0	1	0	1	$100\times20\times100\times1$
0	1	1	0	$20\times100\times50\times20$
0	1	1	1	$20\times20\times50\times1$
1	0	0	0	$20\times20\times200\times100$
1	0	0	1	$20\times50\times200\times20$
1	0	1	0	$50\times20\times10\times100$
1	0	1	1	$50\times50\times10\times20$
1	1	0	0	$20\times20\times100\times20$
1	1	0	1	$20\times50\times100\times1$
1	1	1	0	$50\times20\times50\times20$
1	1	1	1	$50\times50\times50\times1$

What is the most likely configuration?

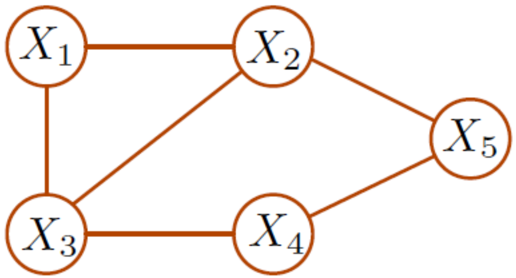
Markov Random Fields

Joint distribution over random variables $X_1, \dots, X_n$ given by:

1. An undirected graph with nodes $X_1, \dots, X_n$ and edges representing dependencies.
2. A distribution that factors over this graph:

$$P(X_1, \dots, X_n) = \frac{1}{Z} \prod_C \psi_C(\{X_i : i \in C\})$$

where the product is over maximal cliques in the graph, and the **clique potentials** ψ_C are positive-valued functions.



Functional form of $P(X_1, X_2, X_3, X_4, X_5)$

$$\frac{1}{Z} \psi_{123}(X_1, X_2, X_3) \psi_{25}(X_2, X_5) \psi_{34}(X_3, X_4) \psi_{45}(X_4, X_5)$$